Fiscal Year:	FY 2023	**Task Last Updated:**	FY 05/28/2023
PI Name:	Vanapalli, Siva		
Project Title:	Determining Muscle Strength in Space-Flown Caenorhabditis elegans		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	79409-0001	**Congressional District:**	19
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	05/19/2015	**End Date:**	05/31/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	4
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 05/31/2022 per NSSC (Ed., 3/3/22) NOTE: End date changed to 12/31/2021 per F. Hernandez/ARC (Ed., 7/27/21) NOTE: End date changed to 9/22/2021 per F. Hernandez/ARC and NSSC information (Ed., 9/16/2020) NOTE: End date changed to 3/31/2021 per F. Hernandez/ARC (Ed., 3/12/2020) NOTE: Extended to 4/30/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/12/19) NOTE: Extended to 8/31/2019 per F. Hernandez/ARC (Ed., 3/5/18) NOTE: Extended to 5/18/2018 per F. Hernandez/ARC and NSSC information (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Blawzdziewicz, Jerzy Ph.D. (Texas Tech University)		
Grant/Contract No.:	NNX15AL16G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	A major impediment to long-duration space travel is loss of muscle mass and strength during spaceflight. Surprisingly, the small worm Caenorhabditis elegans is a good, cost-effective model for studying the effects of spaceflight upon muscle. Our past experiments on Earth and on the International Space Station (ISS) have shown that the expression of key muscle genes that encode members of a muscle attachment complex is reduced in C. elegans during spaceflight. A key open question is if and how these changes in gene expression result in reduced muscle strength in response to spaceflight. While reduced force production is documented for astronauts, it is yet to be demonstrated for C. elegans. To address this critical knowledge gap, we developed a tiny new device that enables continuous measurement of worm muscle forces. Using this device, on Earth, we have shown that defects in muscle attachment complexes lead to decline in worm muscle strength. Motivated by these past flight data and ground results, we propose to (a) demonstrate the flight-readiness of our miniaturized device for quantitative characterization of muscle strength of C. elegans, (b) test our hypothesis, based on past flight data, that muscle strength decreases in flight as a result of altered muscle gene expression in the worm, and (c) quantify muscle strength loss in spaceflight across multiple worm generations. Successful measurements of muscle strength in flight will make C. elegans a much more complete and comprehensive genomic model for muscle physiology, enabling meaningful translation of discoveries made in C. elegans to astronauts. Our proposed studies have important spin out implications for sarcopenia (an urgent public health problem on Earth) and may produce a terrestrial model for testing potential therapeutic interventions for muscle strength decline in flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies have important spin out implications for sarcopenia (an urgent public health problem on Earth) and may produce a terrestrial model for testing potential therapeutic interventions for muscle strength decline in flight.
Task Progress:	A major impediment to long-duration space travel is loss of muscle mass and strength during spaceflight. Surprisingly, the microscopic nematode Caenorhabditis elegans (C.elegans) is a good, cost-effective model for studying the effects of spaceflight upon muscle. Our past experiments on Earth and the International Space Station (ISS) have shown that the expression of key muscle genes that encode members of a muscle attachment complex is reduced in C.elegans during spaceflight. A key open question is if and how these changes in gene expression result in reduced muscle strength in response to spaceflight. While reduced force production is documented for astronauts, it is yet to be demonstrated for C.elegans. To address this critical knowledge gap, we developed a miniaturized device dubbed "NemaFlex" that enables continuous measurement of worm muscle forces based on deflection of soft micropillars. We integrated the NemaFlex device into a worm loading apparatus, so that gravid adult animals of a specific size can be sorted from mixed cultures and evaluated for muscle strength in individual chambers. The spaceflight hardware and worm culture bags were flown as the Micro-16 experiment to the ISS on February 20, 2021, as part of the Northrop Grumman Cygnus NG-15 cargo mission. Experiments were successfully conducted with the crew recording videos of worms deflecting the micropillars in the NemaFlex device. The key results from the spaceflight investigation are: (1) The body diameter of wild-type worms decreased compared to ground controls, potentially due to altered metabolism; (2) The muscle strength of space-adapted worms was lower than that of corresponding worms cultured on ground. Thus, the Micro-16 experiment showed that muscle strength decline occurs in a physiologically relevant animal model, similar to that found in astronauts. The scientific impact of this study is significant as it opens avenues for future investigations to identify drugs that could alleviate the muscle strength decline in space. This spaceflight project contributed to training of more than a dozen personnel, including students and postdocs, on C. elegans, microfluidics and space biology. The funded work resulted in 16 peer-reviewed publications, several student awards, outreach to K-12 students and commercialization of new technologies.
Bibliography Type:	Description: (Last Updated: 10/20/2025)
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